# Problem Statement 1: Original Program

***Road Junction Graph Using Degree Calculation***

A road network consists of several junctions connected by roads. The road network can be represented using an undirected graph, where each junction is represented as a vertex and each road is represented as an edge.

The objective is to design a C++ program that represents the road network using an adjacency list. The program accepts the number of junctions and roads as input and stores the connections between them. Since the graph is undirected, if junction u is connected to junction v, then v is also connected to u.

The program displays the adjacency list of the graph and calculates the degree of each vertex, which represents the number of roads connected to that junction. It also allows the user to identify junctions whose degree is greater than or equal to a given threshold value K.

This program helps in understanding road connectivity and identifying highly connected junctions in a road network.

**Main Operations:**

1. Input the number of junctions and roads.
2. Create an undirected graph using an adjacency list.
3. Display the graph.
4. Calculate the degree of each junction.
5. Filter and display junctions having degree greater than or equal to k.